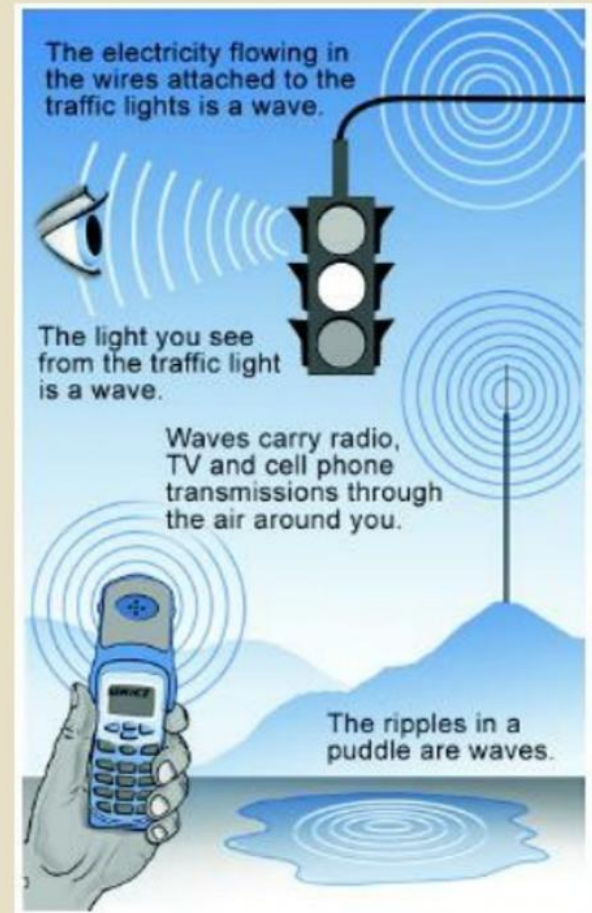


What is a wave?



- Wave- a disturbance that transmits energy through matter or space
- Energy may spread out as a wave travels.
- Waves are all around us:
 - light from the stoplight
 - ripples in a puddle of
 - electricity flowing in wires
 - radio and television and cell phone transmissions



Types of Waves

Waves are mainly classified into three types:

- Mechanical Waves
- Electromagnetic Waves
- Matter Waves

Each type has different properties and modes of propagation.

Mechanical Waves

- Most familiar waves in everyday life
- Governed by Newton's laws of motion
- Require a material medium to propagate
- Examples: Water waves, Sound waves, Seismic waves

Electromagnetic Waves

- Do not require a material medium
- Can travel through vacuum (empty space)
- Examples: Visible light, Radio waves, Microwaves, Ultraviolet rays, X-rays, Radar waves

Speed of Electromagnetic Waves

- All electromagnetic waves travel in vacuum at the speed of light
- $c = 299,792,458 \text{ m/s}$
- Example: Light from stars travels through empty space to reach Earth

Transverse Waves

- A wave on a stretched string is a simple example of a transverse mechanical wave
- Produced by giving one end of the string an up-and-down jerk
- A single pulse travels along the string because the string is under tension

Propagation of Transverse Waves

- When one section of the string moves upward, it pulls the adjacent section upward
- The disturbance continues along the string
- Particles move up and down while the wave travels forward

Longitudinal Waves

- A sound wave can be produced using a piston in an air-filled pipe
- Forward motion of the piston compresses the air
- The disturbance travels along the pipe as a sound pulse

Motion in Longitudinal Waves

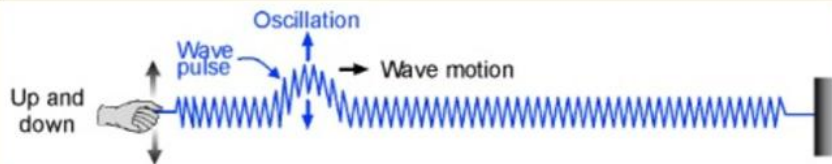
- Backward motion of the piston decreases air pressure
- Air particles move back creating low-pressure regions
- Particle motion is parallel to the direction of wave travel
- Therefore sound waves are longitudinal waves.

Waves

Transverse wave

- Particles oscillate up and down about their equilibrium positions, perpendicular to the direction of wave propagation
- its oscillations perpendicular to the direction the wave moves

Examples: light waves, water waves, all waves belonging to the electromagnetic spectrum



Longitudinal wave

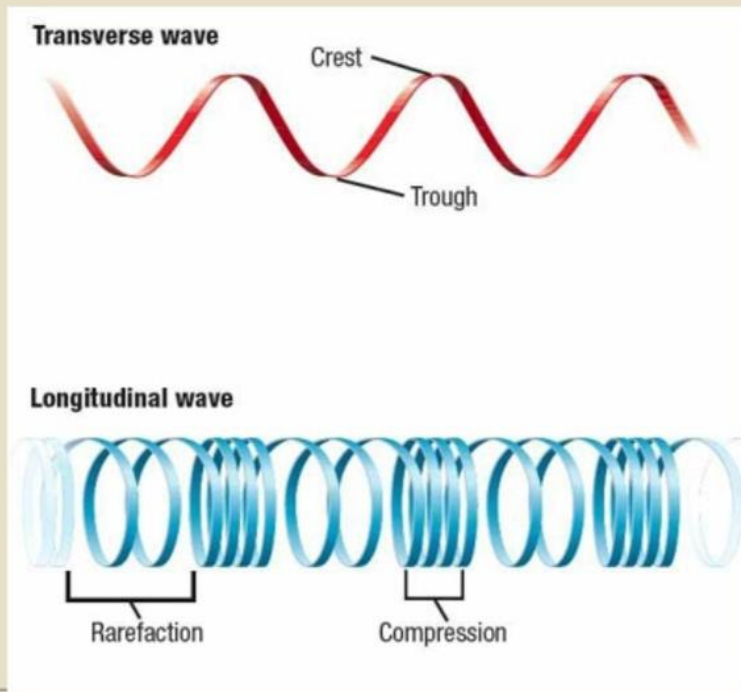
- Particles oscillate back and forth about their equilibrium positions, parallel to the direction of wave motion
- oscillations in the same direction as the wave moves

Example: Sound waves



Waves Properties

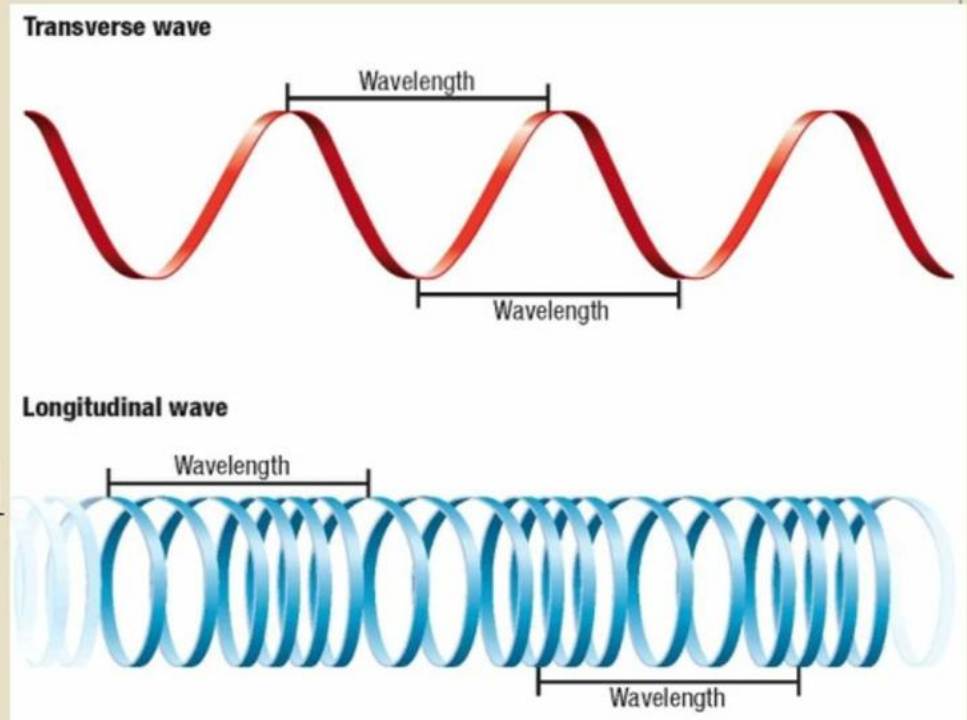
- Waves have crests and troughs or compressions and rarefactions.



- *compressions*: the crowded areas of a longitudinal wave
- *rarefactions*: the stretched-out areas of a longitudinal wave

Wavelength

- Wavelength-the distance between any two successive identical parts of a wave
- represented by the symbol λ
- expressed in the SI unit meters (m)

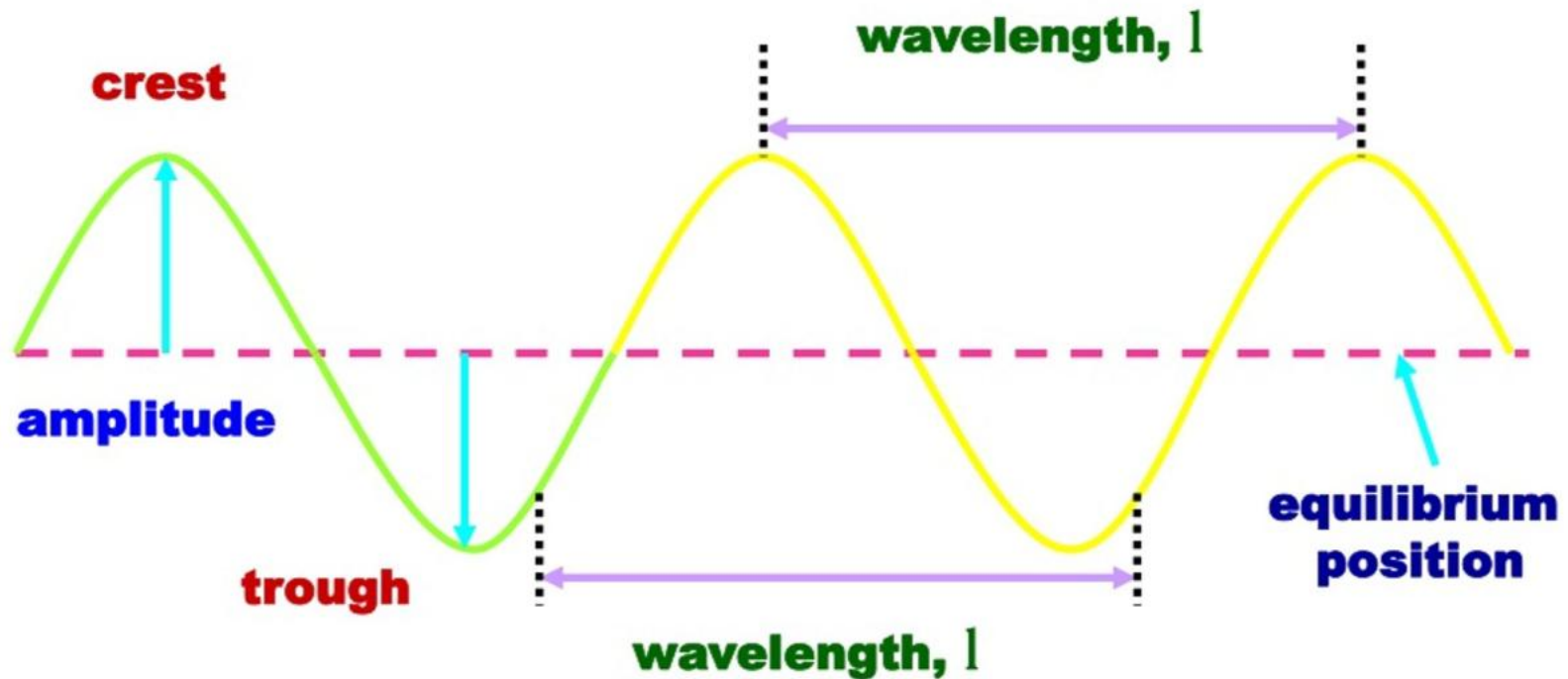


Amplitude and wavelength tell you about energy.

- **larger amplitude = more energy**
- **shorter wavelength = more energy**

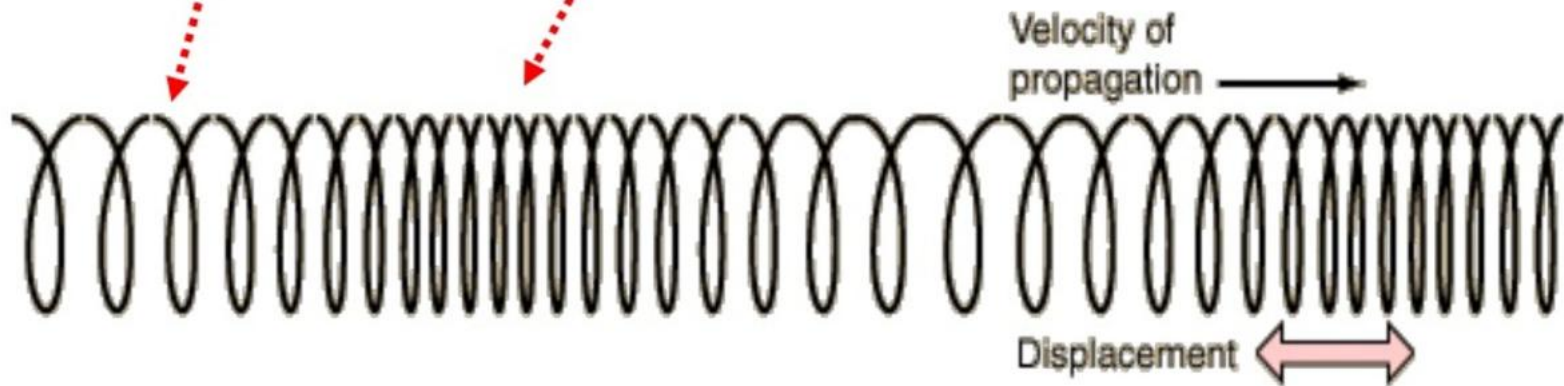
A Transverse Waveform

A **cycle** consists of **one crest** and **one trough**.



A Longitudinal Waveform

- A **cycle** consists of **one compression** and **one expansion** of the particles of the medium.

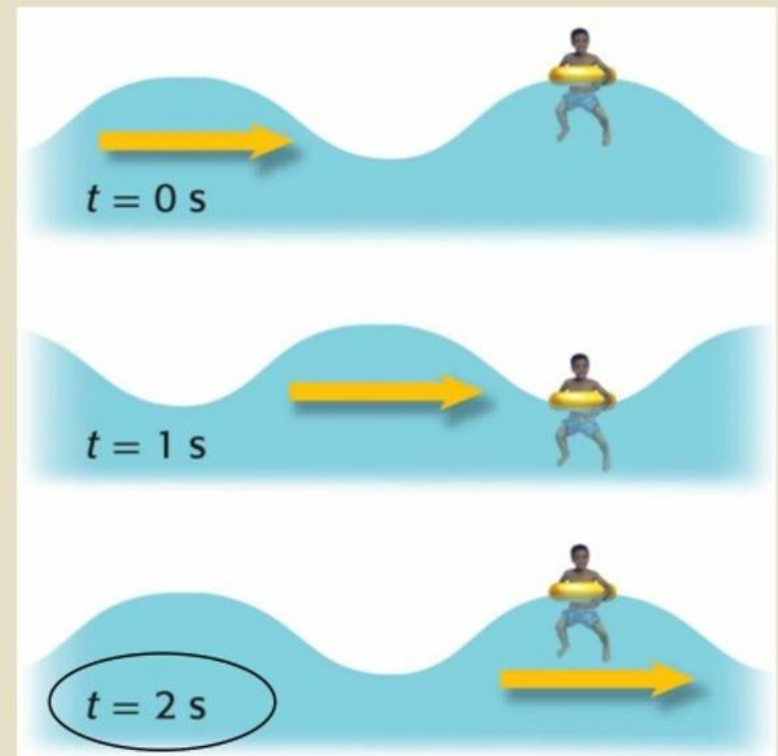


- Particles of medium then return to their **equilibrium positions**

Time Period

Defined as the time required for one full wavelength to pass a certain point

- the time that it takes a complete cycle or wave oscillation to occur
- represented by the symbol T
- expressed in the SI unit seconds (s)



Frequency

- The frequency and time period of a wave are related.
- The frequency is the inverse of the time period.

$$\text{frequency} = \frac{1}{\text{time period}}, \text{ or } f = 1/T$$

**in the diagram,
 $f = 0.5 \text{ Hz}$**

